

Do I Really Feel Better or is it just the Smartwatch Telling Me that I do? - Negotiating the Felt-Experience with Smartwatch Data

Markus Löchtefeld

Department of Architecture, Design and Media Technology, Aalborg University
Aalborg, Denmark
mloc@create.aau.dk

Abstract

Fitness trackers are among the earliest and most accessible forms of human augmentation, integrating seamlessly with the body to monitor, analyze, and optimize physical performance in real time. While previous research mainly focused on how users interpret and respond to the data these devices generate, less attention has been paid to situations where there are discrepancies between the data presented and users' own somatic awareness and felt experiences. This paper addresses that gap through an autoethnographic reflection on using an Apple Watch during recovery from pneumonia, highlighting how mismatches between lived experience and biometric data led to ongoing negotiations with the device. The purpose is to explore the entanglement of personal embodied recovery from acute illness with the technological mediation of health data.

CCS Concepts

• **Human-centered computing** → *Ubiquitous and mobile computing theory, concepts and paradigms*; **Interaction design theory, concepts and paradigms**.

Keywords

wearables, health, apple watch, autoethnography, pneumonia

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1 Introduction

Humans have been developing tools and technologies to improve our capabilities and efficiency for millions of years [23] and similarly HCI research has focused on building advanced tools to open up new possibilities for several decades now [11]. As these tools become increasingly integrated with our bodies, the line between human and machine blurs, deepening our connection to technology. We commonly use terms like augmented human, cyborg, hybrid human, posthuman, and bionic person to refer to this concept of extending our human abilities with novel technologies. Engelbart

outlined already in 1962 a vision for using computers to extend human intellect, proposing systems that enhance our ability to solve complex problems through improved tools and methods of thought [7]. To address the emerging challenges at the intersection of the natural and the artificial, some scholars have begun referring to this convergence as human-computer integration, signaling a shift toward hybrid identities that challenge and redefine traditional notions of what it means to be human [24]. While the augmented human movement — driven by biosensing and sensory augmentation — holds the potential to profoundly reshape human experience, it also raises a host of unresolved questions [34]. As technology continues to evolve at a rapid pace, the human body remains relatively constant, highlighting a growing asymmetry between our biological capacities and the capabilities of the tools we design to extend them.

This transformation often begins with subtle, everyday technologies that gradually become ubiquitous. Fitness trackers are among the earliest and most accessible forms of human augmentation, integrating seamlessly with the body to monitor, analyze, and optimize physical performance in real time. By examining how users interpret and engage with the data these devices generate, such as heart rate variability, sleep patterns, and activity levels, we can gain valuable insights into how technology supports and interacts with human physiology [12, 20, 31]. These insights can inform the design of more advanced augmentation technologies that not only track but also adaptively respond to biological signals, enabling real-time optimization of health, performance, and even cognitive function.

In the past, HCI research has focused mainly on the adoption of and interaction with such devices. For example, how self-tracking goals can emerge from user internal needs and evolve over time [25] or how these tools can be used in citizen science studies [18]. And while we assume that technology and the data it produces are reliable, it is also clear that this is not necessarily the case when users do not use it appropriately [18]. This then inadvertently shapes the narrative their fitness tracker presents, and thereby can lead to misleading interpretations.

Although there is a large body of work that investigates how we make sense of and react to the data produced by smartwatches, fitness trackers, and other smart devices [12, 13, 20, 28], only a few previous studies focused on the possible discrepancy between the data presented to the user and their own somatic sense and felt experience [28, 32]. In this paper I will contribute to this field of work with a reflection based on an autoethnography of using an Apple watch to recover from pneumonia and how the discrepancy between my own lived experiences lead to constant negotiations with the data.



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2 Methodology

In recent years, HCI has increasingly recognized the first-person experience of the researcher as a legitimate and valuable source of knowledge, and we have seen a steady increase in publications based on autoethnographies [4, 16, 21, 26]. This approach is grounded in the belief that exploring the researcher's own emotions, bodily sensations, and thoughts can provide meaningful insights within the research process. Autoethnographies are typically conducted as long-term studies in which researchers immerse themselves in the research environment, either openly or discreetly. Their experiences are documented through fieldnotes, such as photographs, and audio or video recordings. This raw material is afterwards transformed through a reflective writing process that enables deeper analysis and interpretation of their lived experiences [4, 6].

This shift to more qualitative and (auto)ethnographic methods has also led to research on the lived experience of using self-, fitness- and health tracking [1, 13, 19, 28, 32]. It also reflects a deeper transformation in perspective that recognizes that a variety of everyday activities, frequently without a defined goal in mind, can gather novel insights. By moving beyond the optimization-centric view of self-tracking, researchers have opened new avenues to explore how these practices influence and are embedded in the fabric of daily life [8, 27]. In this paper, I follow Forlano's approach [9] to capture, reflect on, and convey the sensorial and emotional depth and complexity of engaging with a technological device during illness (also similar to Homewood [13]).

2.1 Method

In early May 2025, I was diagnosed with pneumonia and had severe breathing problems (particularly shortness of breath and pain while breathing) and sharp stabbing chest pain, especially when breathing deeply and coughing. Pneumonia is a lung infection that causes inflammation in the air sacs of one or both lungs, which can fill with fluid or pus, leading to symptoms such as cough, fever, chills, and difficulty breathing. It can be caused by bacteria, viruses, or fungi, and its severity ranges from mild to life-threatening, especially in vulnerable populations such as the elderly, infants, and those with weakened immune systems [22]. Luckily, I do not belong to any of these populations and, after a week of penicillin-based treatment and another week of recovery time, I was back in full health. This creates some problems: First, of course, this was not a planned autoethnography. I started taking notes after being diagnosed with pneumonia. I started collecting wearable data using an Apple Watch Series 9 three days earlier, prompted by the onset of symptoms that I initially suspected to be cardiac related. This early use was to monitor my heart rate and perform ECG readings to provide reassurance. On top of this, I used it to collect all possible physiological- and sleep data, with data transmission to the Apple Health app for overview. The second problem is that the time period is unusually short for an autoethnography. However, this is often the case with acute illnesses and I tried to make up for it by using a lot of time (since there was nothing else to do) to take audio notes and reflections on the data.

Between February 2019 and March 2024, I consistently used various models of the Apple Watch to track personal data. However, from March 2024 to May 2025, I transitioned to exclusively wearing

a mechanical watch. I also discontinued Apple Watch use entirely 25 days after receiving the diagnosis and completing recovery. In total, I recorded 3 hours 18 minutes of audio notes, including my interpretation of my bodily state and my interpretation of the Apple Watch data. The audio notes were automatically transcribed and synthesized through thematic analysis in two separate sessions. Based on the emerging themes, I synthesized three short vignettes that reflect my experience recovering from pneumonia and the way the data from the Apple Watch supported this and caused confusion.

3 Negotiating Experiences and Measurements

3.1 Vignette 1 - Searching the Heart

It started with a dull ache in my shoulder and a tightness in my chest, subtle but persistent enough to make me feel uneasy - the reason I dugged out the Apple Watch in the first place again. I tapped it, opened the ECG app, and waited for the familiar waveform to scroll across the screen. Normal sinus rhythm. Again. And again. I started obsessively checking heart rate and ECG, as I feared I had had some form of heart attack. I scrolled through the Health App's heart related elements like it might reveal a hidden clue. The data was always clean, reassuring, clinical. But the discomfort lingered and, with it, a growing sense of confusion. If my heart was fine, then what was wrong? At the time, I didn't know I had pneumonia. I was looking in the wrong place, trusting the Apple Watch to tell me what my body already knew but couldn't yet name. The silence of the data made me doubt my own lived sensations, even as the illness quietly deepened.

3.2 Vignette 2 - Breathing in Numbers

The Apple Watch tapped gently against my wrist, its sleek face lighting up with reassuring data in the Blood Oxygen app. It told me that I was doing well: my VO2 max was slowly climbing from 87 to 90, my respiratory rate had normalized, and my heart rate variability suggested resilience. But my body disagreed. Each breath still felt like pulling air through fog. I moved slowly and cautiously as if my lungs collapsed if I moved too fast. In the morning I had woken up drenched in sweat, my chest aching, my energy gone before I even left the bed. Yet the Apple Watch remained optimistic - its metrics clean, its graphs smooth with unwaveringly positive attitude. I began to question the narrative it was telling. The first days when I felt better and stronger, when breathing became easier again, the same data that told me before I was getting better, were still showing signs of stress. The days before, when I felt like I was unraveling, the Watch reported stability. It did not capture fear, fatigue, or subtle changes in my breath that only I could feel. I realized that while the Apple Watch could track my body, it could not fully understand it. Recovery is not just about numbers. It's about how those numbers feel when you are the one living them.

3.3 Vignette 3 - Loose Data

I woke up feeling lighter: my breath was less labored, my body was no longer weighed down by the fog of illness. It was the first morning in days that I didn't dread sitting up. I reached for my phone, eager to see if the Apple Watch had captured the progress I felt. Instead, the Health App greeted me with data painting a different

picture: my blood oxygen had dipped and my respiratory rate had spiked during the night. A red flag in the data, just when I thought I was turning a corner. The panic started to flicker. Had I imagined the improvement? Was my body betraying me again? Then I noticed the strap. The watch had slipped loose during the night, barely making contact with my skin. The data had not captured my body; it had captured a gap, a misreading, a night of disconnection. I tightened the band and laughed, a little breathless, but relieved. The numbers hadn't lied, exactly — they just hadn't known the whole story.

4 Discussion

These three vignettes highlight the most important experiences I had when recovering from pneumonia and are arranged in temporal order not in terms of importance. Vignette 3 for example - while it was an important experience that had a distinct impact on me - reflects a well-understood problem area. Krämer et al. for example reported on how external circumstances such as a cat sleeping on the bed can influence the data, in this case a sleep tracker, to a large degree. Similarly, participants themselves affect the accuracy of the data depending on the complexity of the task and their level of experience [18]. Devices such as the Apple Watch can roughly estimate the uncertainty; for example, if the band is too loose when taking a dedicated blood oxygen measurement, the watch will report that the measurement cannot be taken. This is not the case for continuous monitoring. Also, at no point is the uncertainty or potential ambiguity in the data presented. This lack of ambiguity in the data representations poses a significant problem and creates the idea of objectivity for fitness tracking devices [5]. Although ambiguity has been used as an actionable concept for designers [10], only a few studies have adopted these tactics in relation to fitness tracking devices or the enhancement of human senses [5, 29]. Introducing ambiguity to communicate the uncertainty in the data will be crucial for the future of biosensing technologies. Especially, delivering this in a manner acceptable to users has been a problem in the past. For example, when Strömel et al. used an Large Language Model (LLM) to generate data descriptions of quantitative fitness data [31], tried to remove the ambiguity of the LLM replies after the initial user feedback who did not appreciate it. And while their work showed that the LLM can create a richer, more reflective engagement with personal well-being data, it also hinted at the current issues in designing with ambiguity. Since similar issues also plague current LLMs [17] and other AI systems, there is a potential to learn from this area as well and adapt it to wearable systems. The conclusion of vignette 3 is not necessarily novel, but given the future outlook and integration of AI systems, it is of absolute importance that we continue in the line of Di Lodovico et al. [5] and establish ways of including ambiguity in design in a way that will also be accepted by users.

Vignette 1 and 2 are closely related to each other. And the underlying problem for both is the same: A missing understanding of how to interpret the bodily felt experiences and align them with the measured data. Reflection of the user on fitness data is a cyclic process in which users constantly seek conceptual and temporal alignment with their lived experience [2]. And while temporal alignment was easy for me (given the time on hand to focus solely on recovering),

conceptual alignment at the beginning was problematic. Given that I did not expect to have pneumonia, I did not check my breathing rate or blood oxygen. I was controlling cardiac performance as the initial pain was in that region and these are prevalent in my family. Only after the symptoms became much worse and a doctor diagnosed me, I focused on the right measures.

Roomkham also found in his autoethnography that there is a need to facilitate and help users interpret the raw data by connecting it to their experience and additional knowledge [28]. Tholander and Nylander introduced the concepts of "measured" versus "lived" performance to explore how technology and data influence sports experiences [33]. And Tholander extended this with his subjective, embodied dimensions of performance that go beyond metrics that spark discussions about the role of data in shaping human performance and perception [32]. But so far, few approaches have been aimed at connecting bodily felt experience with measured data.

Höök uses the term soma design to describe a process that incorporates body and movement into the design process of interactive devices [14]. Previous work often focused on specific devices that are designed through a somaesthetic approach to support bodily introspection [14, 30]. However, using data provided by, for example, a fitness tracker could also act as a prompt for somatic sensing and action [32]. Boer et al. presented a diary in which users first estimate and then measure their blood glucose levels, followed by annotating perceived symptoms. Thereby they encouraged the users to reflect on their bodily experiences and symptom patterns in relation to the measurements [3]. Through these kind of interactive approaches we could learn to understand our somatic senses better and at the same time regain the trust in data from a wearable again.

Vignette 2 showcases how I constantly tried to make sense and negotiated between my lived experience and the data produced by the Apple Watch. And these kinds of negotiations have been reported in other previous autoethnographic works [13, 28, 32]. Homewood, who used a Fitbit to limit bodily activity to recover from Long-COVID reported that the *"knowledge produced by my body was seen as less valuable than the data produced by my Fitbit"* [13] and Tholander stated that he had similar negotiations in which lived experience could simply not be captured through the data of his fitness watch [32]. Especially when recommendation systems are coming into play this misalignment of bodily felt experience and the measured data can be quite poignant and feel aggressive. For example, during my recovery period, I found a LinkedIn post depicting a similar problem. Melkumyan describes her experience using a Garmin watch that recommends activities in the following way: *"The problem is: those are often my hardest days. Days when I'm stressed and stuck at a desk. Or juggling things at home. Or caring for a sick kid. The assumption that low heart rate = low stress = easy day doesn't hold."*¹. All of this touches directly on one of the grand challenges of human-computer integration: The effects of these new technologies on identity and behavior [24]. How will users with integrated technology possible shift their self-perception and somatic senses? The alignment of lived experience and feedback from human augmentation devices in health will be important to ensure that we can achieve long-term benefits and not only short-term gains, as is the case with current devices [15].

¹Mariam Melkumyan, May 2025, LinkedIn

5 Conclusion

With fitness trackers being one of the first examples of technology that can be considered to support human sensorial augmentation, it is important to also focus on the lived experience of using these devices. Establishing these can be quite hard in a controlled study, since a variety of everyday activities, frequently without a defined goal in mind, can gather novel insights. That is why the recent increase in the focus on autoethnography is highly important for the field [1, 13, 19, 28, 32]. The autoethnographic study presented in this article contributes to the conversations by documenting and reflecting on personal and subjective accounts of recovering from an acute illness and the interconnection with technology and data. Especially such situations of recovery from a disease like pneumonia would be very hard to cover in another way.

The themes that emerge point towards two important directions that we will have to focus on to ensure that these kinds of negotiations between the lived experiences and the measured data become less of an issue in the future. Firstly, we need to find methods to present ambiguity and uncertainty in ways that are acceptable and useful to users. Secondly, we need to establish interactions that allow the user to reflect on their bodily experiences and symptom patterns in relation to the measurements. It will be key that we align self-perception with the data to not create frustrating or disturbing experiences. We need not only to focus on potential performance improvements or new capabilities. We also need to focus on how these can help us better understand our own "natural" body and our natural lived experiences.

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